Mini-Test

Modern Physics

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Text and solution for Monday's mini-test.

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WEEK VII

- An electron in a hydrogen atom is in an excited state in the 3p/4d subshell with *maximal/minimal* m. What are its quantum numbers? (0.5pt each) *Ignore the electron spin.*
  What is the energy of the electron in Rydberg units [$1\,\mathrm{Ry} = 13.6\,\mathrm{eV}$]? (0.5pt).

  Solution
  The relevant quantum numbers are given in the following:

  $$3p: \quad n = 3 \quad \ell = 1 \quad m = \{-1, 0, 1\} \qquad E_n = -1/9 \text{ Ry}$$

  $$4d: \quad n = 4 \quad \ell = 2 \quad m = \{-2, -1, 0, 1, 2\} \qquad E_n = -1/16 \text{ Ry}$$

  Discussion
  The hydrogen states are identified by their quantum numbers (n, ℓ), where $n = 1, 2, \ldots, \infty$, $\ell = 0, 1, \ldots, n-1$, and the quantum number $|m| \leq \ell$.

  The orbitals corresponding to different angular momentum are labeled: s ($\ell = 0$), p ($\ell = 1$), d ($\ell = 2$), f ($\ell = 3$), and then alphabetically afterwards (g, h, i, ...) based on a historical classification used in spectroscopy to describe groups of lines of atomic spectra.

  The energy of the hydrogen state instead depends only on n as

  $$E_n = -\frac{1}{n^2} E_0,$$

  where the negative energy (bound state) is given in terms of $E_0 = 13.6\,\mathrm{eV}$ or $1\,\mathrm{Ry}$.